Practical No – 9

Nearest Neighbour Analysis, Sampling Raster Data using Points or Polygons & Interpolating Point Data

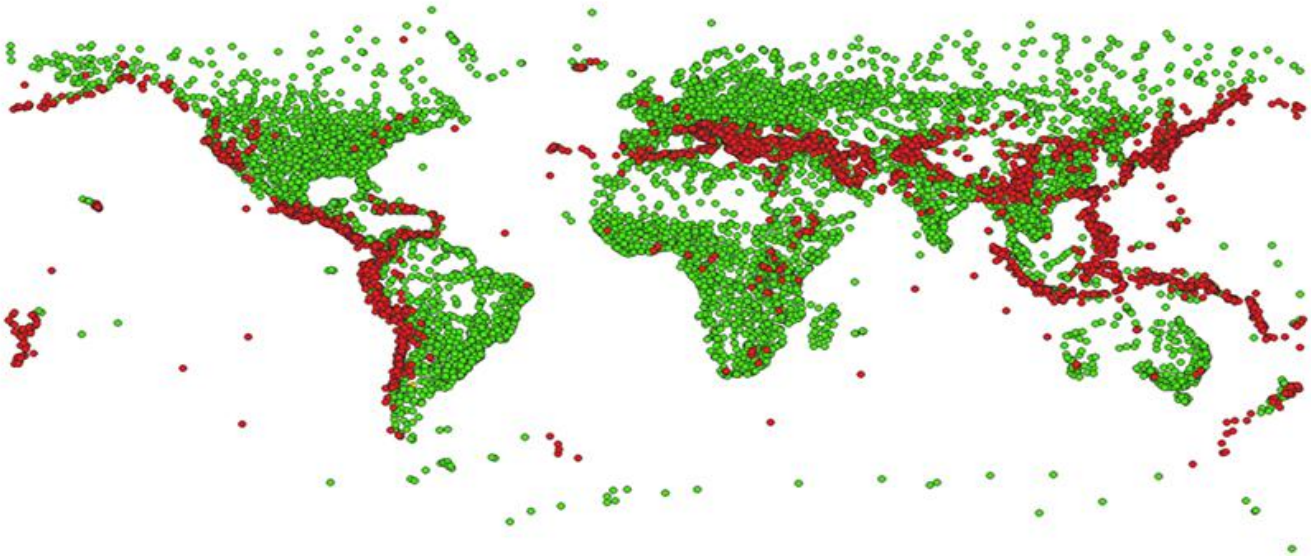
A. Nearest Neighbour Analysis

- Go to Layer→Add Layer→Add Delimited Text Layer and Load

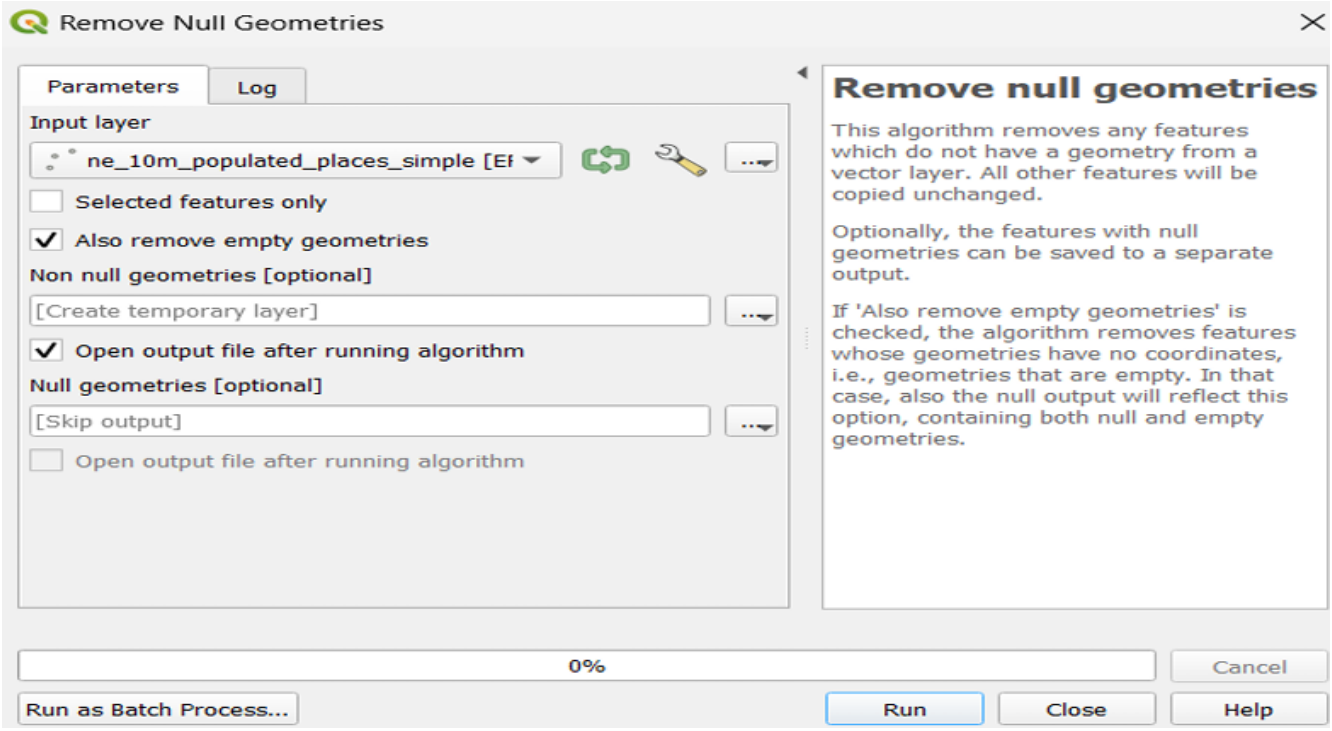
"earthquakes_2021_11_25_14_31_59_+0530.tsv"from data file.

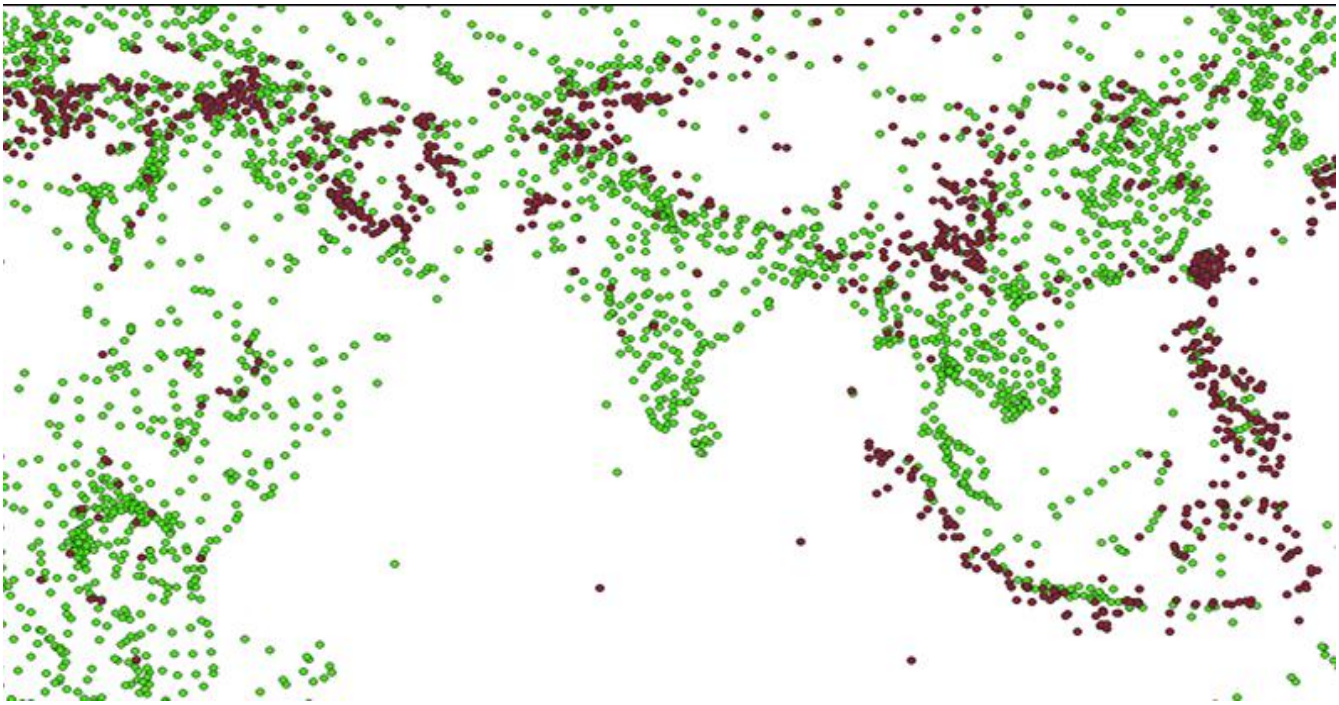
- Go to Layer→Add Layer→Add vector layer and from data folder

"\GIS_Workshop\Practicals\Practical_08\A\DATA\ne_10m_populated_places_simple.zip"

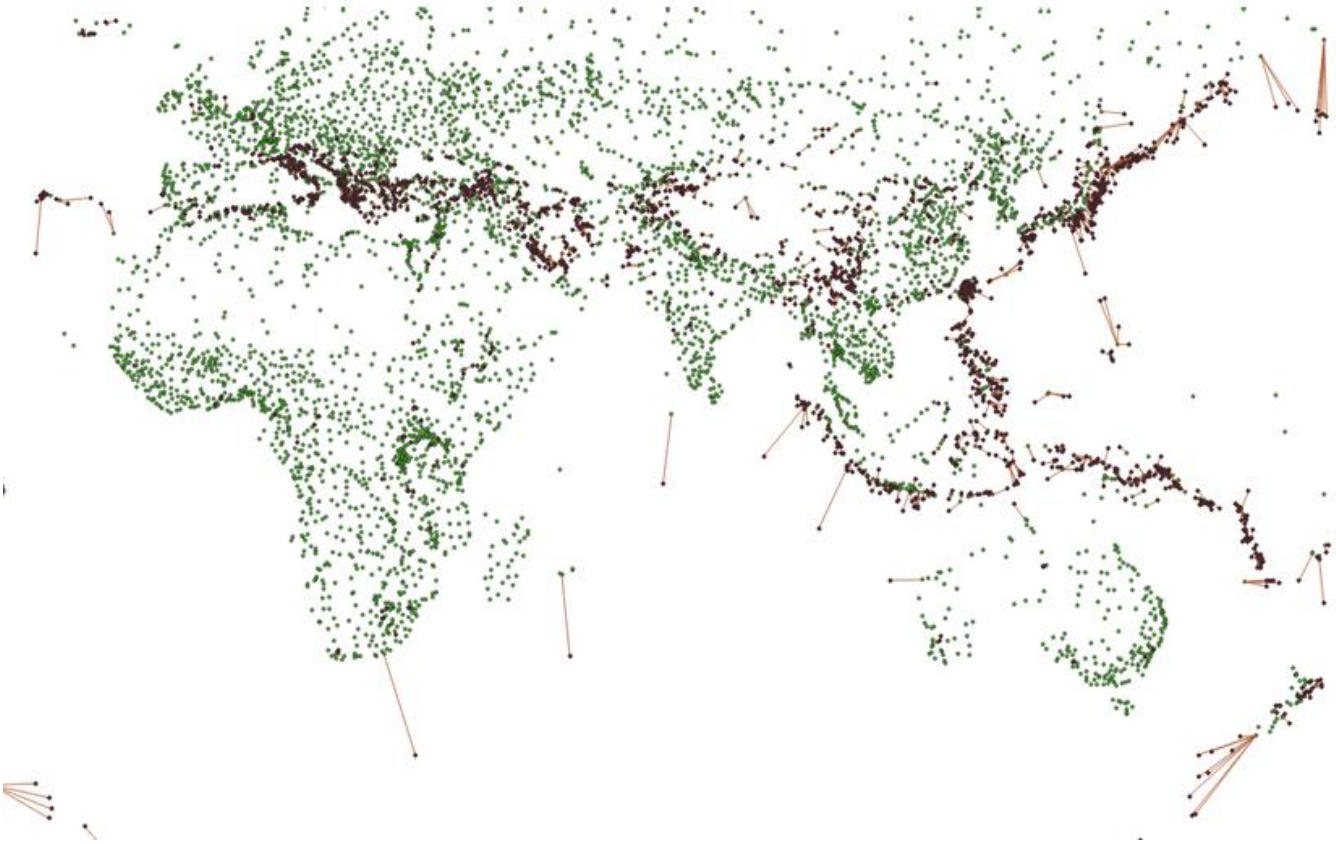


- Go to Processing→Select Toolbox→Search Remove Null Geometries



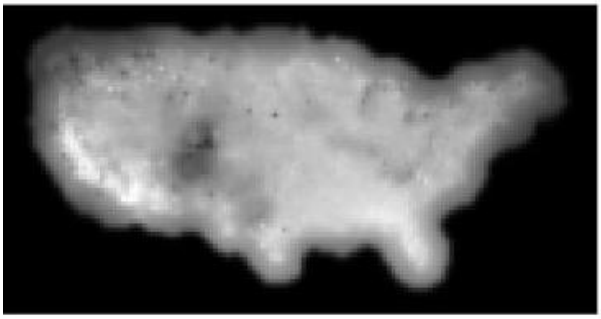


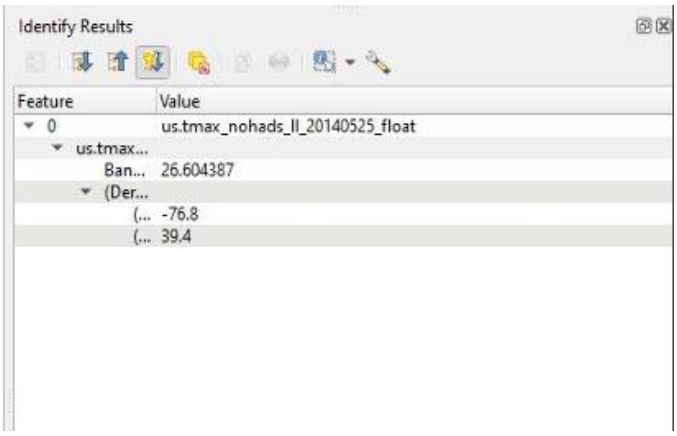
- Go to Processing→Select Toolbox→Search Distance to nearest hub(line to hub)



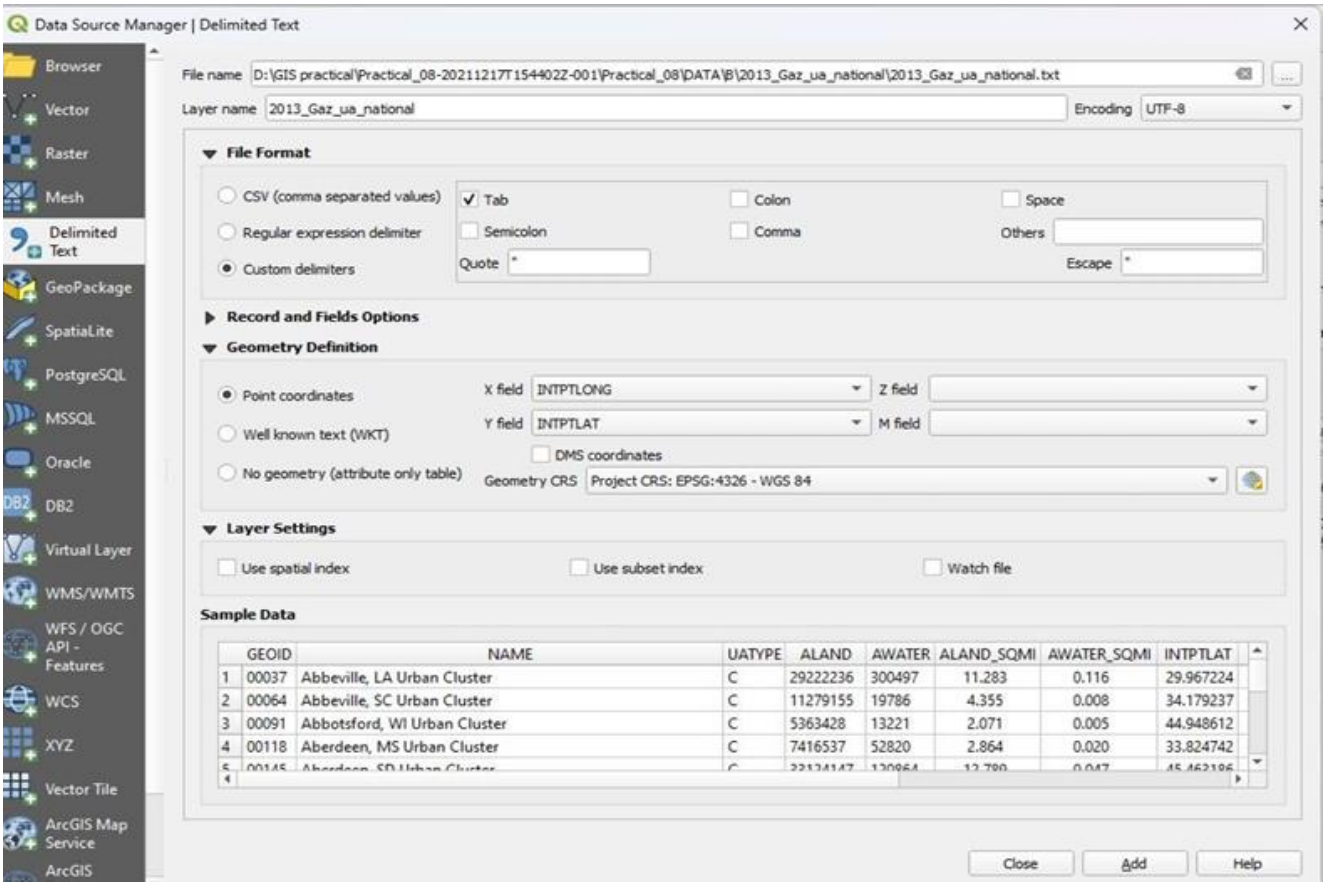
B. Sampling Raster Data using Points or Polygons

- Go to Layer→Add Raster Layer and browse to the downloaded **Us.tmax_nohads_ll_ {YYYYMMDD} _float.tif file** and click **Open.**
- Select the Identify tool and click anywhere on the layer.



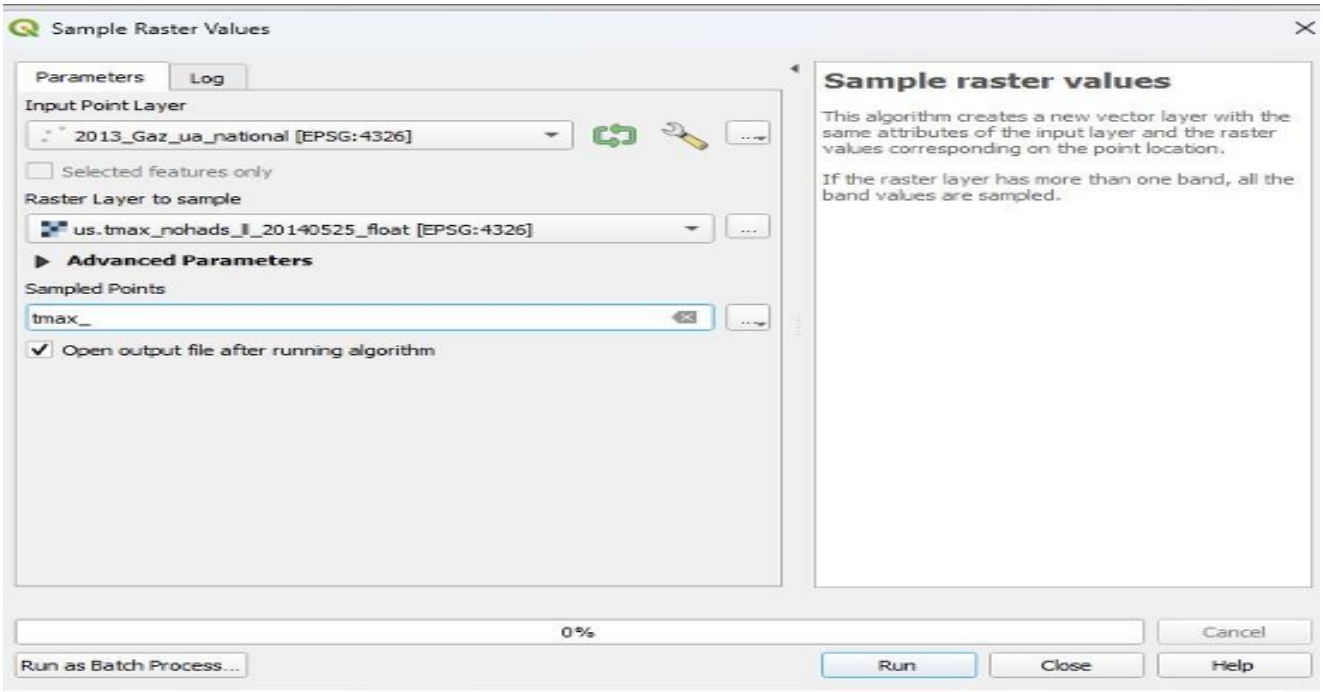


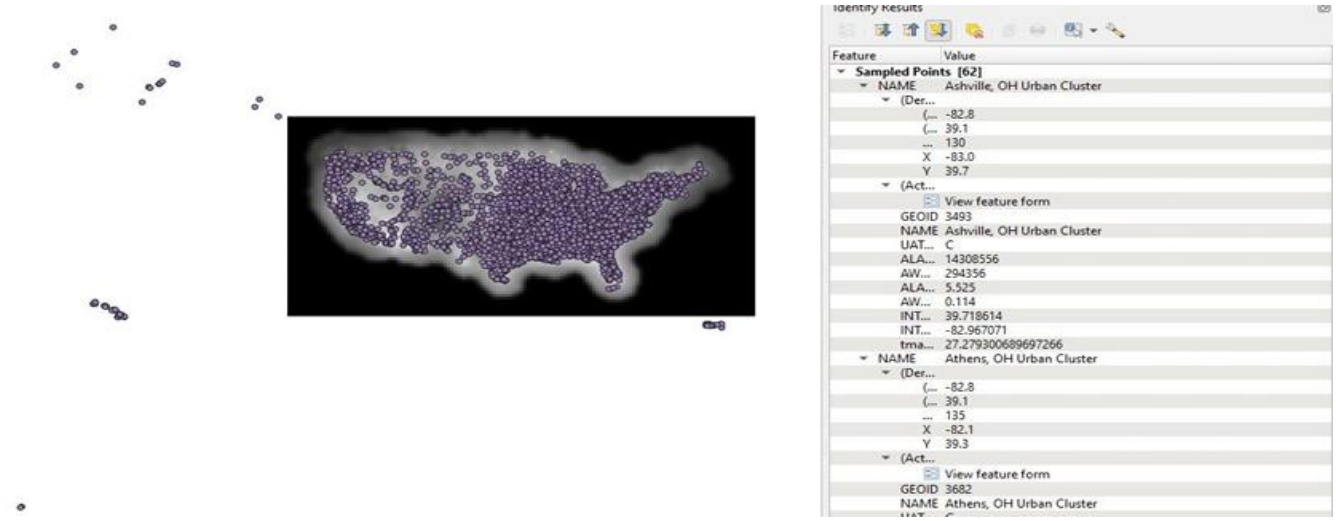
- Now unzip the downloaded **2013_Gaz_ua_national.zip** file and extract the **2013_Gaz_ua_national.txt** file on your **disk.**
- Create a Layer from Delimited Text File Dialog, click **Browse** and open **2013_Gaz_ua_national.txt.**
- Select INTPLONG as X field and INTPLAT as Y field. Check the use spatial index box and click **OK.**



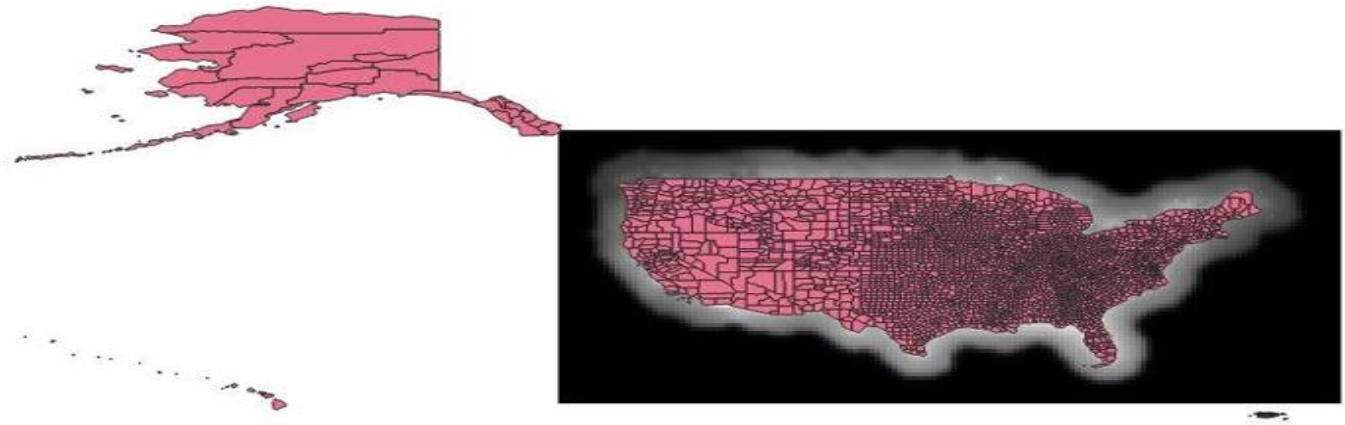
- Install the Point Sampling Tool Plugin.

- Open the plugin dialog from Plugin→Analysis→Point Sampling Tool.

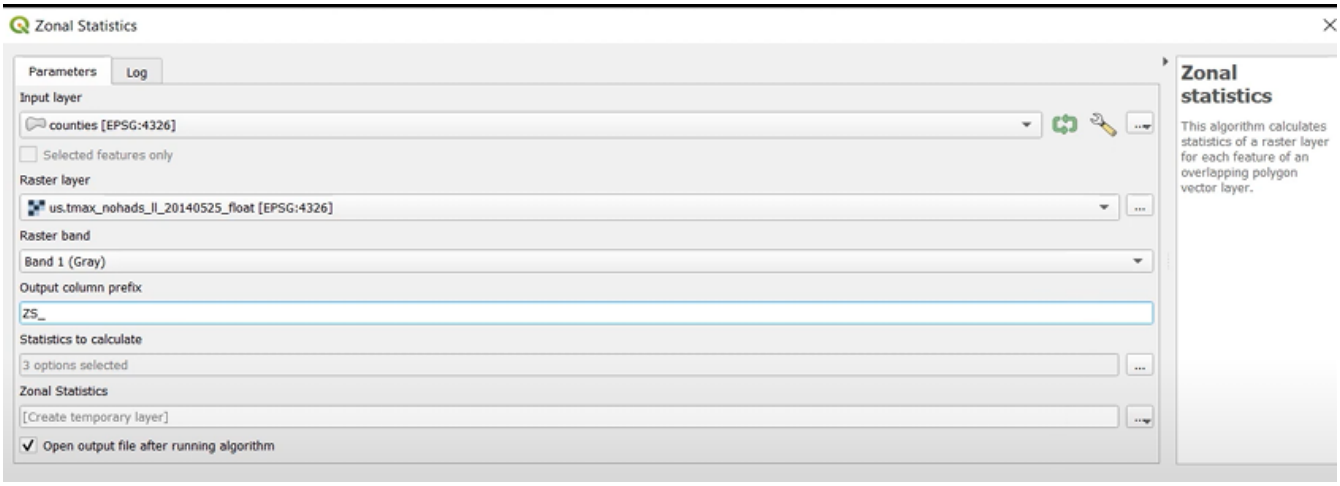


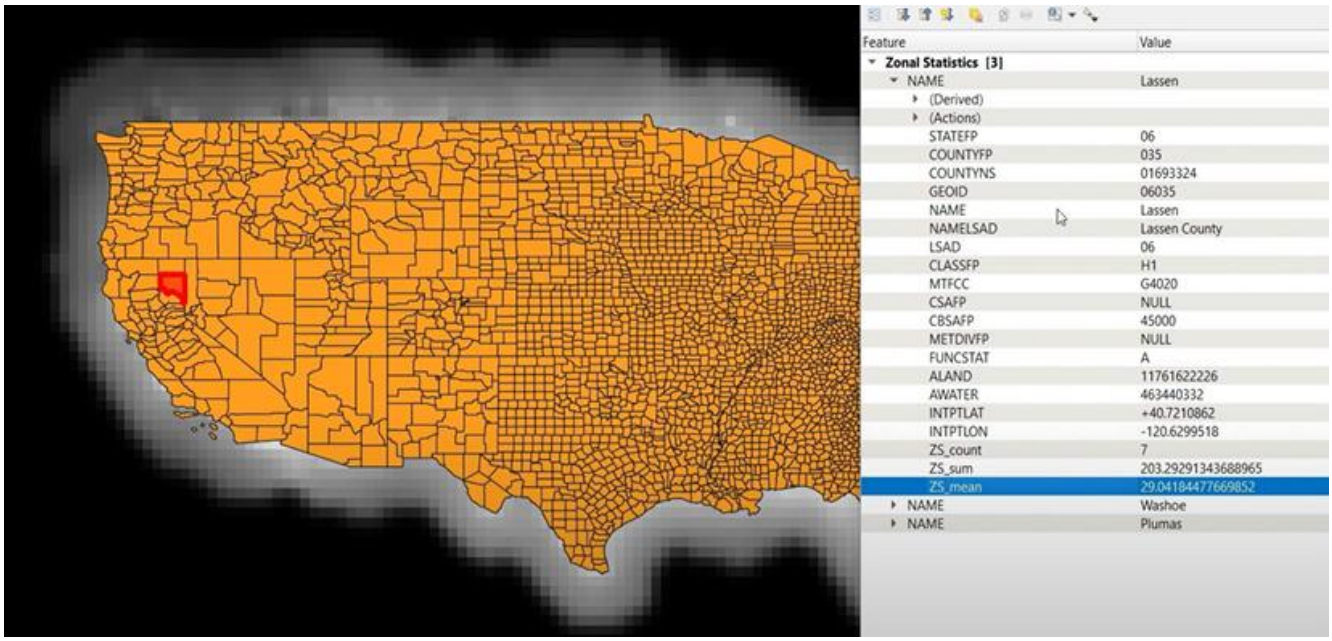


- In the Save Vector layer as... dialog, click Browse and name the output file as counties.shp. Choose Selected **CRS** from the CRS dropdown menu. Click **Browse** and select WGS 84 as the CRS. Check the Add saved file to map and click **OK.**

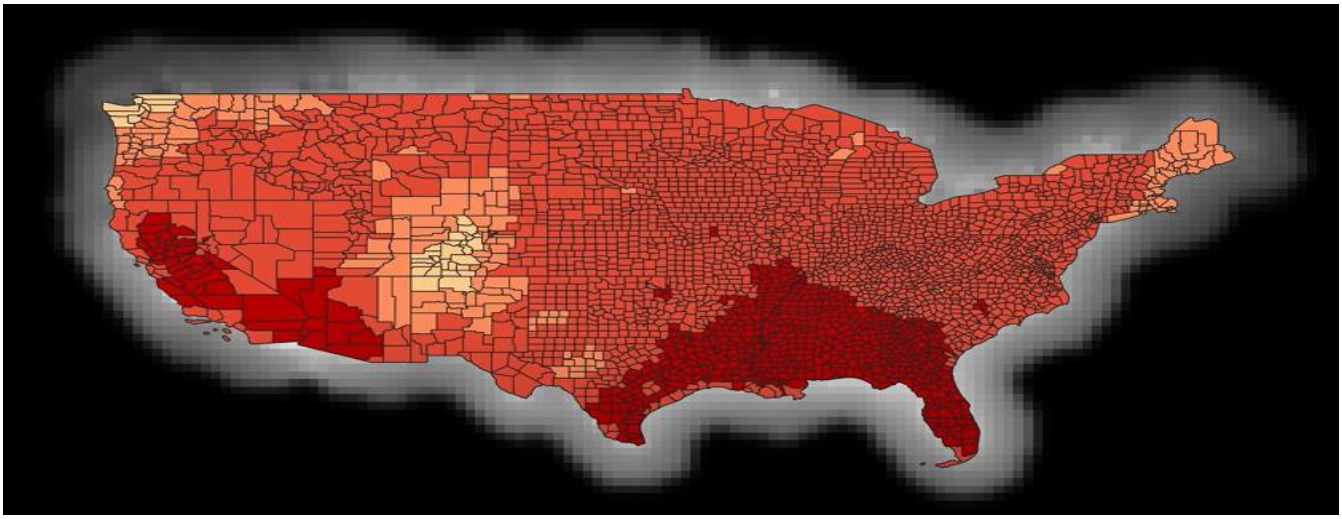


- Enable the Zonal Statistics Plugins. Go to Raster→Zonal statistics→Zonal statistics



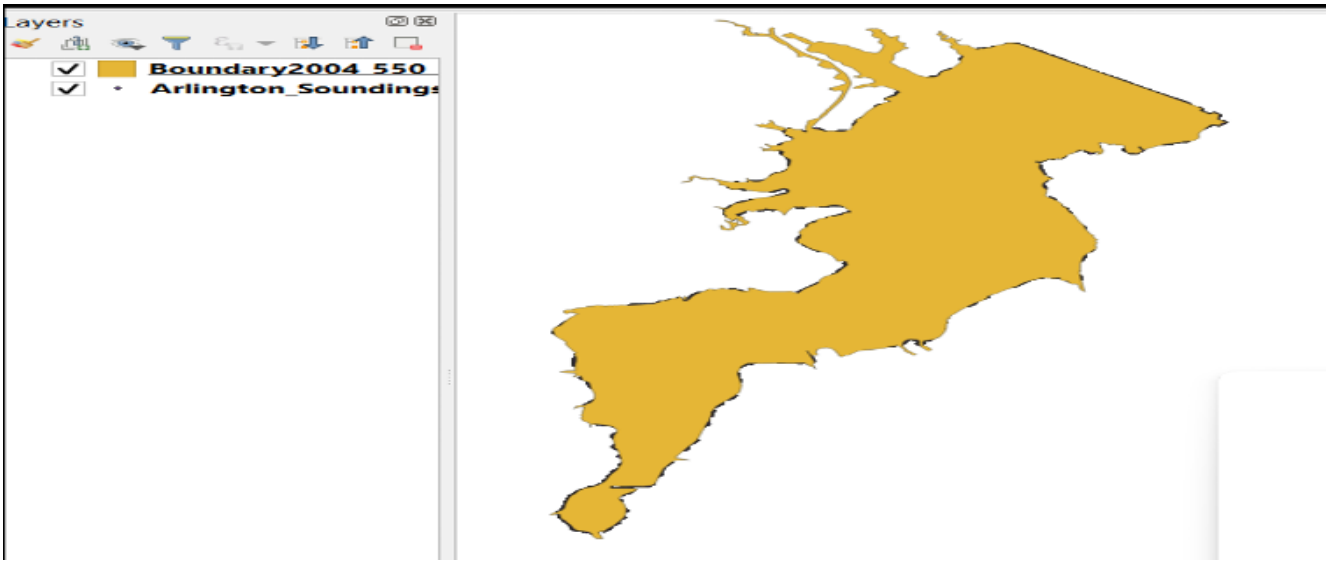


- Right-click the counties layer and select Properties.
- Switch to the **Style** tab. Choose Graduated style and select ZS_mean as the Column. Choose a Color Ramp and Mode of your chose. Click Classify to create the classes. Click **OK.**

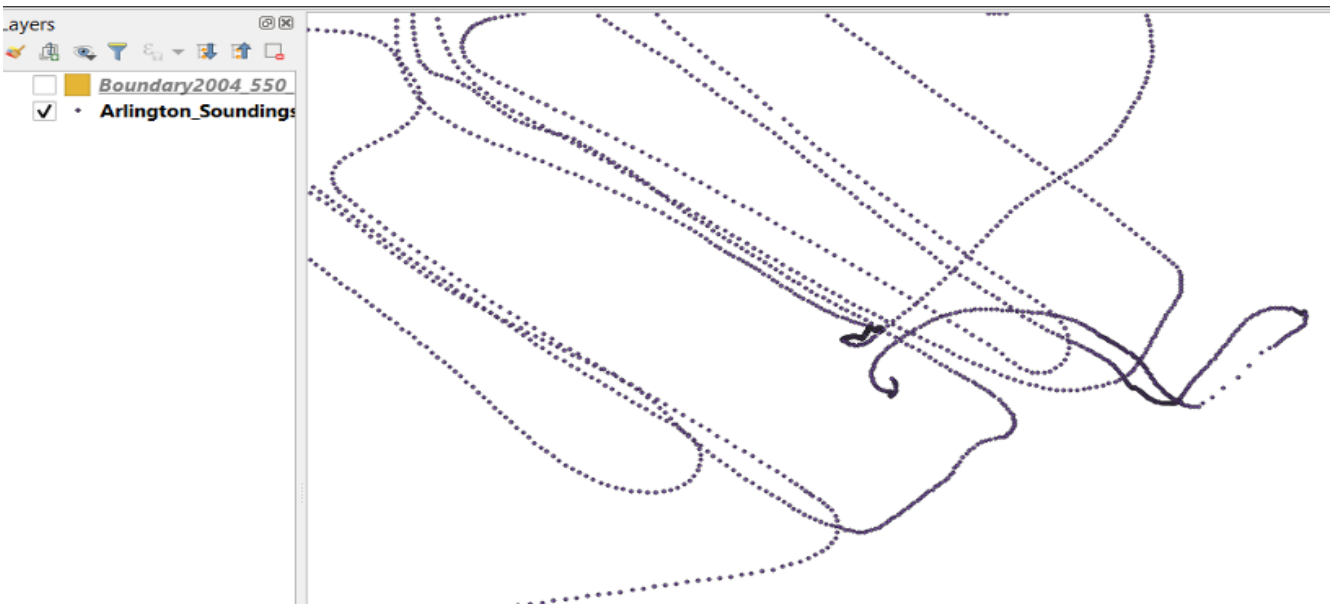


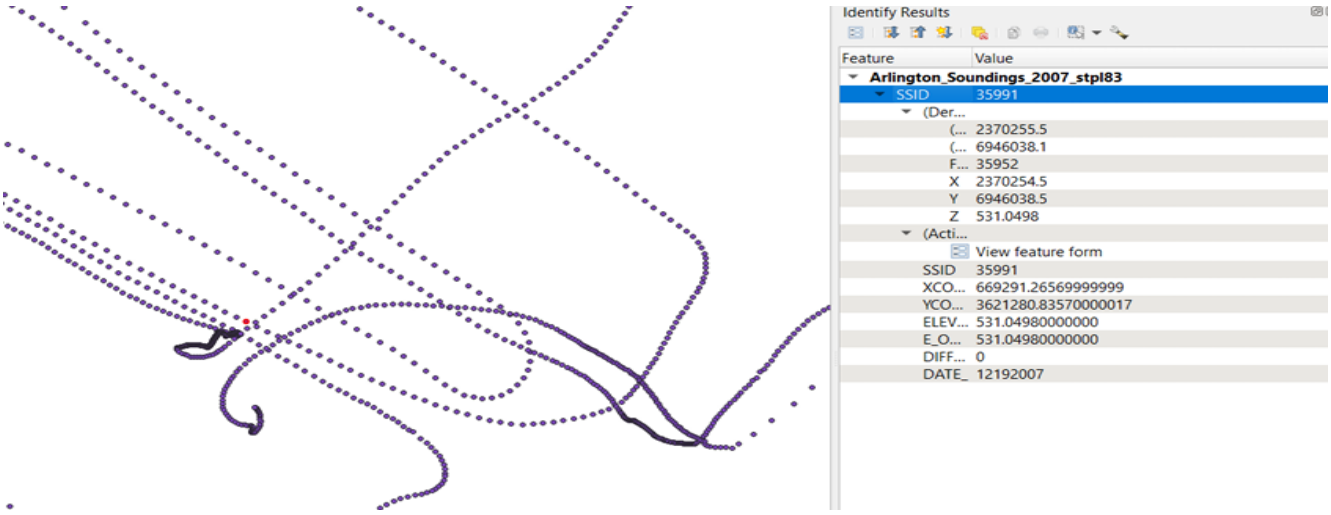
C. Interpolating Point Data

- Go to Layer→Add Layer→Add Vector Layer...
- Browse to the downloaded **Shapefiles.zip** file and select it. Click **Open.**
- Select Arlington_Soundings_2007_stp183.shp and Boundary 2004_550_stpl83.shp layer. Click **OK.**



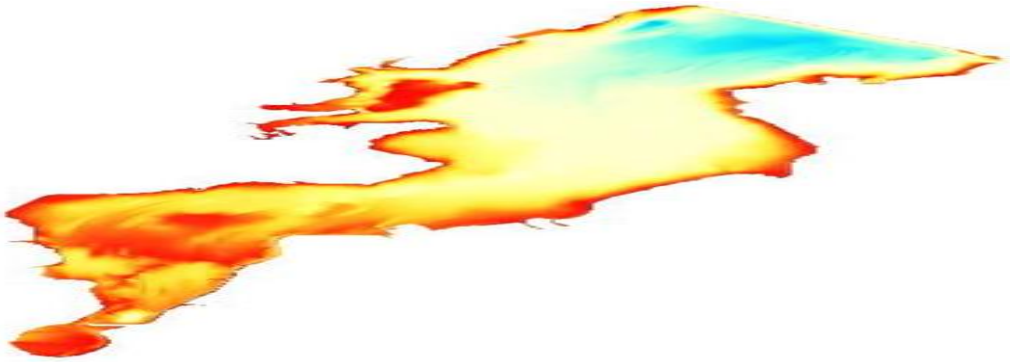
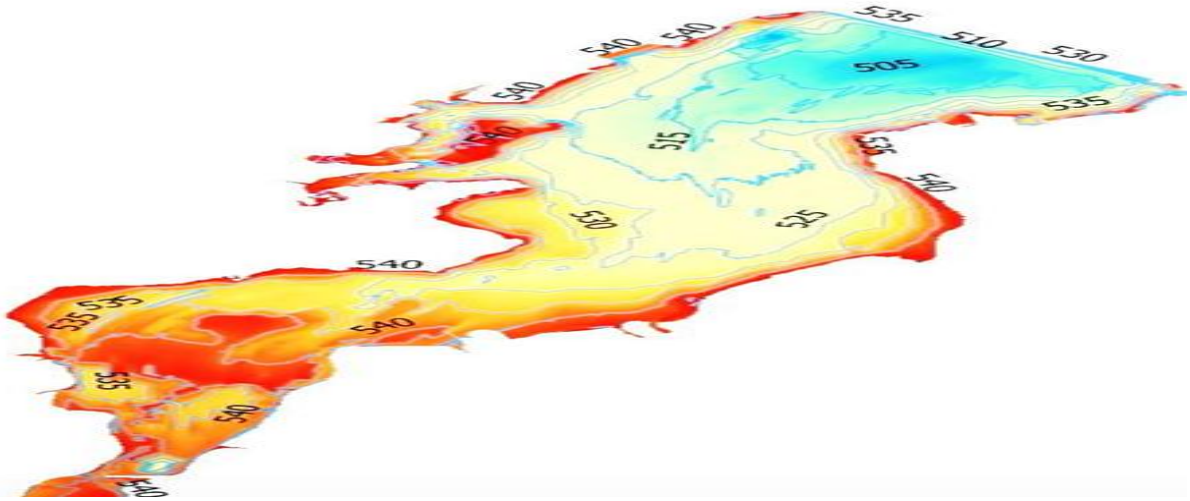
- Click the Zoom icon and select a small area on the screen.





- Go to Raster→Interpolation→Interpolation.
- In the Interpolation dialog, Select Arlington_Soundings_2007_stpl83 as the Vector layers in the Input Panel. Select ELEVATION as the Interpolation attribute. Click Add. Change the Cellsize X and Cellsize Y values to 5.
- Go to Raster→Extraction→Clipper

- The Output file as **elevation_tin_clipped.tif.** Select the clipped mode as Mask Layer. Select **Boundary2004_550_stpl83** as the Mask Layer. Click OK.
- Go to the style tab. Select Render type as Singleband pseudocolor.Click Classify.



- Switch to the Transparency tab. Enter 0→Click **OK**
- Go to Raster→Extraction→Contours. intervals, so enter 5.00 as the Interval between contour lines. Check the Attribute name box. Click OK.